

Team Members: Shaurya Agarwal, Nathan Fireman, Owen Mank, Rajit Mukhopadhyay, Jamil Takieddine
Professor: Dr. Brian Beaudoin | ENEE 408J

INSPIRATION

Bringing **Yondu's arrow** from **Guardians of the Galaxy** to life



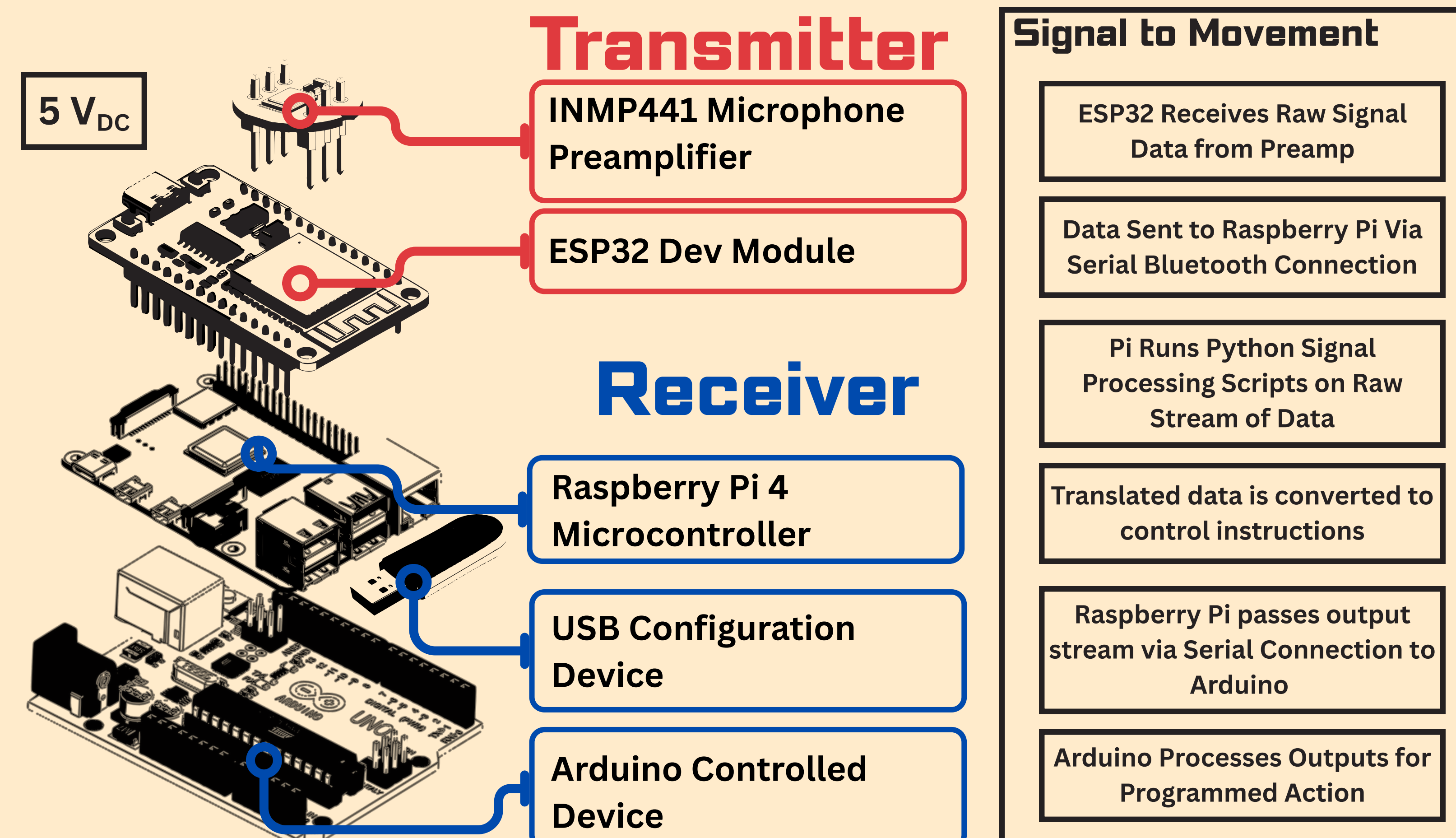
OBJECTIVE

A **configurable system** to **translate complex audio cues** into **independent output instructions** in the audible range for **sensing and control**

APPLICATIONS

- **Smart Homes (Consumer Tech):** Tone-based triggers to control lights, appliances or locks
- **Factories (Industrial Automation):** Systems can self-disengage if frequencies associated with mechanical failure are detected
- **Interactive Exhibits (Arts & Education):** Allows visitors or students to trigger displays using musical notes

ARROW DESIGN



SIGNAL PROCESSING

1. **Convert Signal** from **I²S** to **dB SPL**
2. **Bandpass Filter** to audible spectrum
3. **Subtract Noise Floor** to **improve signal clarity**
4. **FFT** to isolate signal frequency peaks
5. **Peak Grouping** to **reduce processing complexity**

CONFIGURATION

JSON Format:

GLOBAL VARIABLES

Configuration Name
Signal Sensitivity (Hz)
Amplitude Sensitivity (dB)

SIGNAL - OUTPUT MAP

Frequency Peak (Hz)
Frequency Sensitivity (Hz)
Amplitude Sensitivity (dB)
Output Code (String/Char)

EXAMPLE:

```
{
  "config_name": "two_tone_test",
  "Global_Sensitivity": 10.0,
  "Global_Amplitude": 65.0,
  "Peaks": {
    "Peak1": {
      "frequency": 440.0,
      "sensitivity": 5.0,
      "output_code": "A"
    },
    "Peak2": {
      "frequency": 523.25,
      "sensitivity": 5.0,
      "output_code": "C"
    }
  }
}
```

DEMO

Watch here:



... or try it yourself!

ANALYSIS

- **Electrical Noise Reduction:** Reducing the length of wire used and soldering our connections minimized electrical noise.
- **Constructive and Destructive Interference:** In its current state, signal clarity is impacted when notes are played simultaneously.
- **Note Testing and Identification:** Once frequency detection was consistent, we recorded the frequency ranges generated by striking varying notes on a tongue drum. After this we identified 4 notes with fairly unique frequencies.
- **Testing Applications of Sensor:** Connecting our Raspberry Pi to a Lafvin Arduino bot, we programed it to drive forward or backward and to turn left or right based on which of the 4 notes it received.

